

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)

November – December 2012

IT 202 Data Communication

Date: 01.12.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Do as Direct.

- (a) A broadcast radio transmitter radiates 10 kW when the modulation percentage is 60. [02]
How much of this is carrier power?
- (b) Calculate the output signal – to – noise ratio in decibels for three links, the first two of [03]
which have S/N ratios of 60dB and the third an S/N of 40dB.
- (c) Define transformation ratio in tapped inductor. [01]
- (d) Define Passive Circuit. [01]

- Q - 2 (a) Draw the block diagram of superhetrodyne receiver and show the output at various [08]
points.
- (b) Define the term insertion loss as applied to attenuators and explain T attenuator with [06]
necessary equation.

OR

- Q - 2 (a) Give the difference between series tuned and parallel tuned circuit. [06]
- (b) Write a note on sensitivity and gain of receiver. [08]
- Q - 3 (a) Write short note on Partition noise. [04]
- (b) Define amplitude modulation and modulation index. Use a sketch of a sinusoidally [05]
modulated AM waveform to help explain the definition.
- (c) Write short note on RC Phase Shift oscillator. [05]

OR

- Q - 3 (a) Write short note on Flicker noise. [04]
- (b) Derive the relation between the output power of an AM transmitter and the depth of [05]
modulation, and plot it as a graph for values of the modulation index from zero to
maximum.
- (c) Write short note on Voltage – controlled oscillator. [05]

SECTION – II

Q – 4 Do as Direct

- (a) Define Noise Figure. [01]
- (b) List the different type of tracking of superheterodyne receiver and which is best suitable? [02]
- (c) The tuned circuit of the oscillator in a simple AM transmitter employs a 50 – microhenry coil and a 1 – nanofarad capacitor. If the oscillator output is modulated by audio frequencies up to 10kHz, what is the frequency range occupied by the sidebands? [03]
- (d) Popcorn Noise is also called as a _____. [01]

- Q – 5 (a) Write short note on different properties of Fourier transform. [06]
- (b) Draw the generalized hybrid model for BJT and derive its following parameter. [08]
1. current gain
 2. input impedance
 3. voltage gain
 4. output impedance

OR

- Q – 5 (a) Draw the hybrid model for three different configuration of bipolar junction transistor. [08]
- (b) What is periodic and aperiodic signal? Compare power and energy signals. [06]
- Q – 6 (a) Which type of feedback is used in oscillator and why? [02]
- (b) Give the difference between ideal filter and practical filter. [02]
- (c) A coil has a series resistance of 5 ohm, a self-capacitance of 7 pF, and an inductance of 1 micro – henry. Determine the effective inductance and effective Q – factor when the coil forms part of a series tuned circuit resonant at 25 MHz. [03]
- (d) Explain diode envelope detector with necessary circuit diagram and waveforms. [03]
- (e) A receiver has a noise figure of 12 dB, and it is fed by a low noise amplifier that has a gain of 50 dB and a noise temperature of 90 K. calculate the noise temperature of the receiver and the overall noise temperature of the receiving system. [04]

OR

- Q – 6 (a) Describe energy spectral density with its properties. [02]
- (b) What is frequency synthesizer? Give its different type and also give the disadvantage of PLL frequency synthesizer [02]
- (c) Explain skin effect. [03]
- (d) Draw the block diagram of AM broadcast transmitter. [03]
- (e) Calculate the noise factor of an attenuator pad that has an insertion loss of 6 dB. [04]
